

PolyBLEP Anti-Aliasing Residual Polynomials

Band-Limited Synthesis Math for [osc~] and [phasor~] Objects

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Abstract

This document formulates the 2-point PolyBLEP (Polynomial Band-Limited Step) residual function equations utilized in **Time Dilation DAW** to suppress high-frequency aliasing foldback during non-bandlimited waveform generation.

1 The Aliasing Problem in Naive Digital Oscillators

A naive digital sawtooth waveform $\phi(t) \in [0, 1)$ generated at normalized frequency $dt = \frac{f}{f_s}$ contains discontinuous step transitions at phase wrap points:

$$x_{\text{naive}}(t) = 2\phi(t) - 1.0 \quad (1)$$

By Fourier analysis, a ideal continuous sawtooth contains infinite harmonics decreasing as $\frac{1}{k}$. In discrete sample domain, harmonics exceeding the Nyquist limit $f_{\text{Nyquist}} = \frac{f_s}{2}$ fold back into the audible spectrum, causing harsh metallic aliasing noise.

2 The PolyBLEP Residual Correction Function

PolyBLEP smooths the sharp step discontinuity by subtracting a 2nd-order polynomial residual $\text{blep}(t, dt)$ within a ± 1 sample neighborhood of the phase wrap transition.

The normalized phase offset $t = \frac{\phi}{dt}$ near transition is evaluated as:

$$\text{blep}(t, dt) = \begin{cases} 2t - t^2 - 1.0 = -(1 - t)^2, & 0 \leq t < 1 \quad (\text{Just after step}) \\ t^2 + 2t + 1.0 = (t + 1)^2, & -1 < t < 0 \quad (\text{Just before step}) \\ 0.0, & \text{otherwise} \end{cases} \quad (2)$$

3 Application to Waveform Types

3.1 Band-Limited Sawtooth Waveform ([phasor~], [osc~])

Subtracting the PolyBLEP residual from the naive sawtooth:

$$y_{\text{saw}}(t) = (2\phi(t) - 1.0) - \text{blep}(\phi(t), dt) \quad (3)$$

3.2 Band-Limited Square / Pulse Waveform ([osc~])

A square wave contains two step discontinuities per period: one at $\phi = 0$ (rising edge) and one at $\phi = 0.5$ (falling edge):

$$y_{\text{square}}(t) = y_{\text{naive}}(t) + \text{blep}(\phi(t), dt) - \text{blep}((\phi(t) - 0.5) \pmod{1}, dt) \quad (4)$$

3.3 Band-Limited Triangle Waveform ([osc~])

Integrating a band-limited square wave yields a smooth triangle wave with attenuation of aliasing by over 80 dB across all fundamental frequencies up to f_{Nyquist} .